GRAND ROUNDS CASE · FOLLOW-UP PROGRAM

# Followup 26-06

Five hours of unsteadiness in a hypertensive patient

| | |
|---|---|
| **Patient** | 83-year-old female. Right total hip arthroplasty in May, hypertension, premature ventricular contractions, severe obstructive sleep apnea, chronic kidney disease stage 3a. Prescribed nebivolol 2.5 mg nightly and had stopped taking it about five weeks earlier. Lives alone. |
| **Outcome** | Survived, and never had the thing everyone was looking for. Stroke workup negative on CT, CT angiography, and MRI. Echocardiogram normal with an ejection fraction of 65%. Two days of telemetry found no arrhythmia. **Final diagnosis was orthostatic hypotension.** |
| **Disposition** | Nebivolol permanently discontinued, fludrocortisone started, discharged home on hospital day 2 walking independently, with cardiology follow-up and an ambulatory monitor arranged. |

## The call

Dispatched to a residence for dizziness. She was found supine in bed, alert and oriented times four, airway patent, respirations even and unlabored, skin adequate, lungs clear.

Her son reported an arrhythmia diagnosed in May and a hip replacement the same month. He had noticed extra tablets left in her prescription, and she confirmed she had not been taking the medication for about a month. She had spent the day at the county fair and then mowed the lawn until about 2130, and told the crew she had not had much to drink and suspected dehydration.

The lightheadedness started around 2230. It was still there five hours later, which is why the call was made. **It was present lying flat and worse with position changes**, and it made walking difficult. She denied nausea, vomiting, chest pain, shortness of breath, abdominal pain, or recent illness. BEFAST negative, pupils equal and reactive. Minor bilateral pedal edema since June with no pain, redness, or temperature change. No falls, no head injury, no syncope, no loss of consciousness.

She had felt something like this after her hip surgery in May, but that time it came with dark spots in her vision and a low blood pressure, and she was clear that this was different.

| From contact | BP | MAP | Pulse | Other |
|---|---|---|---|---|
| +7 min | 150/60 | 90 | 70 | Respirations 14 |
| +18 min | 165/84 | 111 | 67 | SpO2 99, glucose 100 |
| +27 min | 160/72 | 101 | 66 | SpO2 99 |
| +37 min | 153/67 | 96 | 64 | SpO2 98 |

**She asked to walk to the stretcher with assistance and did it successfully, and told the crew her lightheadedness did not get worse.** She remained hypertensive and above her baseline through transport, and reported gradual improvement en route.

Two things to hold onto. **She was hypertensive at every single reading**, and **her heart rate never moved.**

## Primary impression

Lightheadedness, most likely from dehydration, in a patient off her cardiac medication.

That is a reasonable reading of what was in front of them, and the patient herself offered the same explanation. The final answer was close to it without being the same thing, and the difference is the whole lesson.

She did have a volume problem, and the labs later proved it. But volume depletion was not the diagnosis. **The diagnosis was orthostatic hypotension**, made by a cardiology clinician who took a blood pressure lying down and again standing up, twenty-nine hours after the crew arrived. Supine 169/73. Standing 137/73. A thirty-two point systolic drop, in a woman who never recorded a systolic below 137 in the entire chart.

The other half of the impression turned out to point the wrong way. Everyone reasonably assumed the missed nebivolol was part of the problem. The workup concluded the opposite, and the drug was permanently discontinued.

## Hospital course

**Emergency department.** Concern documented for arrhythmia versus electrolyte abnormality versus dehydration, with low concern for stroke. Basic labs, magnesium, thyroid studies, ECG, and chest radiograph ordered, plus a 500 mL lactated Ringer's bolus. Labs showed no significant electrolyte abnormality, magnesium and thyroid normal, urinalysis unremarkable, hemoglobin 12.0. ECG showed normal sinus rhythm at 62, with the premature ventricular complexes seen on a prior tracing now absent. Chest radiograph showed hyperinflation consistent with COPD and no acute process.

She was still unsteady on a second attempt to ambulate. Because she lives alone, discharge was judged unsafe, and CT head and CT angiography of the head and neck were obtained and were negative for acute vascular pathology. She was admitted.

**Hospital day 1.** Admitted for stroke rule out, with the working impression documented as most likely a transient ischemic attack, on the reasoning that she had no vertigo and her labs did not suggest hypovolemia. MRI of the brain, echocardiogram, orthostatic vitals, and cardiac monitoring were ordered, and the nebivolol was held. MRI showed no acute intracranial abnormality. Echocardiogram showed a normal-sized left ventricle, an ejection fraction of 65%, normal diastolic function, mild tricuspid regurgitation and no other valvular disease. Telemetry ran normal sinus rhythm averaging 60 to 65.

**Hospital day 2.** Symptoms recurred walking to the bathroom in the morning. Cardiology was consulted for recurrent presyncope and **performed orthostatic vital signs at the bedside: 169/73 supine, 137/73 standing.** The patient stated that immediately on standing she felt she could tip over. Cardiology recommended discontinuing the nebivolol permanently because of the orthostatic hypotension, starting fludrocortisone 0.1 mg daily, and counseled slow position changes, compression stockings, and hydration.

Physical therapy screened her that afternoon, found her independent with mobility and walking about 300 feet with a steady step-through gait, and ran Dix-Hallpike and horizontal canal testing that reproduced no symptoms. She was discharged home the same evening with her son, with cardiology follow-up and an ambulatory monitor to be arranged.

**Final diagnoses:** gait instability from orthostatic hypotension; primary hypertension; premature ventricular contractions; chronic kidney disease stage 3a; obstructive sleep apnea.

# Education points

## Orthostatic hypotension is a change, not a number

This is the single most useful thing in this case, and it is counterintuitive enough that it is worth stating flatly. **A patient can be hypertensive and orthostatic at the same time, and this patient was both, all night.**

The definition has nothing to do with how low the pressure gets. It is a drop of at least 20 mmHg systolic, or at least 10 mmHg diastolic, within three minutes of standing from lying flat. Hers dropped 32 points systolic, from 169 to 137. Both of those numbers are hypertensive. She met the criteria comfortably and never once had a low blood pressure recorded anywhere in the record.

The physiology is why. Standing up drops roughly half a liter of blood into the veins of the legs and abdomen within seconds. Less blood returns to the heart, so the heart has less to eject, so pressure falls. In a working system the **baroreceptors** in the carotid sinus and aortic arch sense that fall within a beat or two and trigger a reflex: heart rate up, contractility up, arterioles constricted. Pressure is restored before the patient notices anything.

What that reflex defends is **the change**, not any particular value. A patient whose usual pressure is 170 has a brain accustomed to being perfused at 170. Drop that person to 137 in ten seconds and their cerebral perfusion falls just as sharply as a patient going from 120 to 88, and they will feel exactly the same thing. Chronic hypertension actually makes it worse, because the vessels that autoregulate cerebral blood flow reset upward over years and lose the ability to compensate at pressures a healthy person tolerates easily.

> **Further reading on orthostatic hypotension.** The neurogenic and non-neurogenic causes and the medication contributors go further than this document can carry. StatPearls has a free chapter with the diagnostic criteria and the full differential: https://www.ncbi.nlm.nih.gov/books/NBK448192/

## The heart rate that never moved

Look at the pulse column again: 70, 67, 66, 64. Then 64 at triage, 69 on the floor, 57 at the cardiology bedside, and telemetry averaging 60 to 65 for two days.

That is a heart rate that does not respond to anything. She had spent a hot day outdoors and mowed a large lawn, she told the crew she had not been drinking much, and her labs later confirmed she was dry. **A volume-depleted 83-year-old should be running faster than that**, and she had been off her beta blocker for five weeks, so there was nothing pharmacologic holding the rate down. That last point rests entirely on the medication history taken at the bedside, since nobody drew a level. The crew asked, the son produced the leftover tablets, and the patient confirmed it, and everything this section concludes is built on that answer.

That flatness is the finding. It says the baroreflex is not producing the response it should, which is the same failure that produces the blood pressure drop on standing. Aging alone does a lot of this: baroreceptors stiffen, the reflex arc slows, and the heart becomes less responsive to sympathetic drive. Add mild volume depletion and a hot day and the reserve runs out.

**A heart rate that does not change when the situation calls for it is a positive finding, not an absence of one.** Tachycardia is easy to notice. The absence of tachycardia in a patient who should have it takes deliberate looking, and it is often the more important observation.

### Prehospital considerations

**This history earns a call for Echo.** A known arrhythmia diagnosed three months ago, five weeks off the medication prescribed for it, and five hours of continuous symptoms that will not resolve is a picture worth putting a monitor on. Not because anybody expected to find something, and in fact nothing was found across two days of telemetry. The value is in the looking. A rhythm strip recorded while a patient is actively symptomatic answers a question that no test done afterward can answer, and a twelve lead is a permanent record of what her conduction looked like during the event rather than after it. **Escalate on the history, not on how the patient looks**, because this patient looked fine.

**You already did an orthostatic test, so say so.** Standing an 83-year-old who is symptomatic lying flat and cannot walk unaided is a real fall risk, and nobody should be criticized for not doing formal orthostatic vitals here. But she asked to walk to the stretcher, she did it with assistance, and the crew documented that her lightheadedness did not get worse. That is a supervised functional test of exactly the thing that turned out to be wrong with her, and it belongs in the verbal report in those words.

**Dizziness is four different complaints sharing one word**, and separating them changes the differential completely. **Vertigo** is the room spinning and points at the inner ear or the posterior circulation. **Presyncope** is the about-to-black-out feeling and points at perfusion. **Disequilibrium** is unsteadiness without any head sensation and points at gait, proprioception, or the cerebellum. **Nonspecific** is everything else. This crew's narrative did that work: no room spinning, present while supine, worse with position change, negative BEFAST, no syncope. That description survived into a chart where the story shifted several times, and it is what a receiving clinician needs.

## The interventions

#### PREHOSPITAL INTERVENTIONS

**Assessment and history.** The most valuable thing done on this call was talking. The son producing leftover tablets, the confirmed five weeks off the medication, the day at the fair, the mowing until 2130, the low fluid intake, the exact onset time, and the patient's own comparison to her post-operative episode in May are all details that exist because somebody was standing in the house asking. Several of them shifted or vanished in later versions of the history, and the account taken at the bedside at four in the morning is the only version of that evening anybody will ever have.

**Blood glucose, 100.** Hypoglycemia is the reversible cause of altered mental status and unsteadiness that you can find and fix immediately, and it gets checked on every one of these.

**Serial vital signs, four sets over forty minutes.** The trend is the reason this document can teach anything. Four blood pressures with mean arterial pressures and four heart rates, spaced ten minutes apart, is what turns a single reading into a pattern, and the pattern here is what points at the baroreflex.

**Assisted ambulation to the stretcher.** Covered above. She asked, it was supervised, it was successful, and the result was documented.

#### IN-HOSPITAL INTERVENTIONS

**500 mL lactated Ringer's.** A balanced crystalloid, closer to plasma than normal saline in its chloride content. Given for the suspected volume depletion, and the labs before and after show it worked, which is covered below.

**CT, CT angiography, and MRI.** A stroke workup on a patient with unsteadiness, no focal deficit, and an entirely normal neurologic exam. Worth understanding rather than second-guessing, because posterior circulation strokes are the ones that present without weakness or facial droop, and they can present as pure unsteadiness. Ruling that out first is the correct order of operations even when the answer turns out to be something else.

**Echocardiogram and two days of telemetry.** Both looking for a cardiac cause of recurrent presyncope: a structural problem on the echo, a rhythm problem on the monitor. Both were negative, and negative results are what allowed the team to stop looking at the heart and start looking at the reflex.

**Orthostatic vital signs.** The test that made the diagnosis, and it costs a cuff and three minutes. Ordered on day 1 and performed at the bedside by cardiology on day 2. The technique matters: supine for at least five minutes, then a reading, then standing, with repeat readings within three minutes.

**Nebivolol stopped, fludrocortisone started.** Nebivolol is a beta blocker, so it blunts the heart rate response that defends pressure on standing, which makes it the wrong drug in a patient with orthostatic hypotension. It was discontinued permanently. **Fludrocortisone** is a synthetic mineralocorticoid: it acts on the kidney to retain sodium, and water follows sodium, so it expands plasma volume and gives the circulation more to work with when the reflex is not doing its share. It is a treatment for the mechanism rather than for the symptom.

## The labs, and what they mean

*Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.*

| Lab | Day 0 | Day 2 | Reference range |
|---|---|---|---|
| Blood urea nitrogen (mg/dL) | 26 | 17 | 8 to 23 |
| Creatinine (mg/dL) | 0.96 | 0.90 | 0.50 to 1.00 |
| eGFR (mL/min/1.73 m2) | 59 | 63 | 60 or greater |
| Sodium (mmol/L) | 137 | 141 | 135 to 145 |
| Potassium (mmol/L) | 3.5 | 4.4 | 3.5 to 5.1 |
| Hemoglobin (g/dL) | 12.0 | 12.4 | 12.0 to 16.0 |
| Neutrophils, absolute (x10^9/L) | not reported | 1.07 | 1.80 to 7.80 |
| Alkaline phosphatase (U/L) | 119 | 105 | 35 to 104 |
| Magnesium (mg/dL) | 2.3 | not repeated | 1.7 to 2.2 |

**The single most useful number here is one the panel does not print, and you calculate it in your head.** Divide the blood urea nitrogen by the creatinine. On arrival that is 26 over 0.96, a ratio of **27**. The next morning, after 500 mL of fluid, it is 17 over 0.90, a ratio of **19**.

Both of those substances are cleared by the kidney, but the kidney handles them differently. Creatinine is filtered and essentially not reabsorbed, so it tracks filtration alone. Urea is filtered and then partially reabsorbed, and the amount reabsorbed rises when the kidney is trying to hold onto water, because urea is reabsorbed alongside sodium and water. So when a patient is dry, urea climbs while creatinine stays put, and the ratio widens. **A ratio above about 20 with a normal creatinine is the fingerprint of volume depletion**, and it is one of the few lab findings you can calculate from two numbers on any basic panel.

The emergency department note reads that there were no significant electrolyte abnormalities and kidney function was at baseline, and the admitting note reasoned that her labs did not indicate hypovolemia. Both statements are true about the individual values. The ratio between two of them told a different story, and it corrected overnight with half a liter of fluid, which is about as clean a confirmation as this kind of reasoning gets.

To be clear about what that does and does not mean: **her volume status was a real contributor and not the diagnosis.** She was still orthostatic on day 2 after fluid, which is exactly why the answer turned out to be the reflex rather than the tank.

**Two values worth reading on your own.** Her alkaline phosphatase ran 119 and then 105, mildly above range on both draws, which was noted for outpatient follow-up. Alkaline phosphatase comes from liver and bone, and in an 83-year-old with a hip replacement a few months earlier, bone turnover is a reasonable place to start looking. Her **absolute neutrophil count on day 2 was 1.07** against a floor of 1.80, which is a mild neutropenia in a patient with a completely normal total white count of 4.1. Neither needed action.

## Learning points

1. **Orthostatic hypotension is defined by the drop, not the destination.** At least 20 mmHg systolic or 10 mmHg diastolic within three minutes of standing. This patient dropped 32 points, from 169 to 137, and both numbers are hypertensive. She was never once recorded with a low blood pressure. If you are waiting for a number below 90 to think about orthostasis, you will miss the patients who have it.
2. **A heart rate that will not move is a finding.** Hers sat between 57 and 70 across two days, in a woman who had spent a hot day outdoors, mowed a lawn, drunk very little, and taken no beta blocker in five weeks. That flatness is baroreflex failure showing itself, and it is the same failure that dropped her pressure on standing. Look for the tachycardia that should be there and is not.
3. **Escalate on the history, not on the presentation.** A known arrhythmia, five weeks off the medication, and five hours of continuous symptoms is worth a monitor and a twelve lead, and worth asking for Echo. Two days of telemetry found nothing, and asking was still right, because a rhythm strip taken while a patient is symptomatic answers a question that nothing done later can answer.
4. **Say what the word dizzy means to this patient.** Vertigo, presyncope, disequilibrium, and nonspecific are four different complaints with four different differentials. This crew documented no room spinning, present while lying flat, worse with position change, and no syncope, and that description was more precise than several of the versions that followed it.
5. **Divide the BUN by the creatinine.** Above about 20 with a normal creatinine means the kidney is holding onto water, which means the patient is dry. Hers was 27 on arrival and 19 after half a liter of fluid. Two numbers on a basic panel, one division, and a finding that three separate notes described as unremarkable.

---

*Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at [https://bcmca.github.io/followup/](https://bcmca.github.io/followup/)*